Design and Implementation of a Web-Based Stock Scoring Engine

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February 19, 2026

References

[1] Majid Behzadian, S Khanmohammadi Otaghsara, Mortaza Yazdani, and Joshua Ignatius. A state-of-the-art survey of vikor methodology. *International Journal of Information Technology & Decision Making*, 11(01):139–181, 2012.

> This research provides a detailed survey of the VIKOR methodology, which is a multi-criteria decision-making algorithm used to reach a compromise ranking from a set of conflicting criteria. The paper explains how the method determines weightings for various qualitative and quantitative inputs to reach a final multi-dimensional score. The authors demonstrate that this framework remains reliable even when data points have completely different scales or units, which is a common challenge in financial data analysis. As a peer-reviewed survey in a leading decision-making journal, this source is essential for justifying the ranking logic used in the 13-metric engine. It provides a mathematical basis for aggregating diverse ratios—like P/E ratios and debt levels—into a singular, objective stock rank. By following the standardization techniques in this study, the scoring engine can ensure that no single metric skews the final results. This ensures the S&P 500 comparison dashboard remains accurate and mathematically sound.

[2] Xiaodan Dong, Weidong Huang, and Jitong Wang. Financial big data visualization: A machine learning perspective. In *Proceedings of the 17th International Symposium on Visual Information Communication and Interaction (VINCI '24)*, New York, NY, USA, 2024. ACM. Accessed: 2026-01-27.

> Visualizing data for over 500 companies is a major challenge for my web app. Dong, Huang, and Wang propose a machine learning approach to solving this Big Data visualization problem. They break down the process into three stages: before, during, and after the model runs. This is useful for me because I need to show a simple Score out of 130, but also allow users to look into the details if they want . I will use their "post-model" visualization strategies to decide how to display the summary statistics and trends for the top 10 stocks in my ranked

list. This ensures that the dashboard remains readable even when processing the full S&P 500 dataset.

[3] Leonardo dos Santos Pinheiro and Mark Dras. Stock market prediction with deep learning: A character-based neural language model for event-based trading. In *Proceedings of the Australasian Language Technology Association Workshop 2017*, pages 6–15, 2017. Accessed: 2026-02-19.

This research investigates the effectiveness of character-level neural language models for forecasting stock market movements based on financial news events. While traditional word-level models often fail to capture the nuances of specialized financial terminology, this character-based approach handles out-of-vocabulary terms more effectively by focusing on morphological features. The methodology employs recurrent neural networks to identify patterns in unstructured text that correlate with both intraday and interday price changes. As a peer-reviewed study presented at the Australasian Language Technology Association Workshop, the source is a highly credible academic reference for deep learning applications in finance. The findings are directly applicable to the News/Catalyst scoring logic within the web-based stock engine. By utilizing character-level analysis, the system can more accurately convert qualitative data into the numerical scores required for the 13-metric ranking system. This ensures that subjective metrics, such as company leadership or market sentiment, are evaluated through a mathematically rigorous framework rather than simple keyword matching. Implementing these deep learning strategies allows the dashboard to remain technically sound while providing users with automated, data-driven insights into news-related stock volatility.

[4] Michael Ettredge, Vernon J. Richardson, and Susan Scholz. A web site design model for financial information. *Communications of the ACM*, 44(11):51–55, 2001. Accessed: 2026-01-27.

Ettredge, Richardson, and Scholz investigate how financial websites should be structured to help investors actually make decisions. They distinguish between hard data such as stock prices and soft information like investor relations updates. This distinction is exactly what I will be dealing with in my project. My 13-metric system combines hard numbers like Revenue Growth with subjective scores like Moat or Leadership, so their design model gives me a way to organize these differently on the screen. Although the paper is older, the core design principles for financial data haven't changed much. I plan to use their ideas to design my Ranked Table so users can see the important metrics without getting overwhelmed by the subjective ones.

[5] Wes McKinney. Data structures for statistical computing in python. In Stéfan van der Walt and Jarrod Millman, editors, *Proceedings of the 9th Python in Science Conference*, pages 56–61, 2010. Accessed: 2026-01-27.

This paper introduces "pandas," the fundamental Python library I will be needing to build my application. McKinney explains the design of the DataFrame, which is the exact data structure I will need to store and rank the 500 stocks. I cited this because my project is heavily dependent on this tool for calculating the 13 metrics efficiently.

[6] Oluwayemisi Runsewe, Lucy Anthony Akwawa, Samuel Olaoluwa Folorunsho, and Olajide Soji Osundare. Optimizing user interface and user experience in financial applications: A review of techniques and technologies. *World Journal of Advanced Research and Reviews*, 23(03):934–942, 2024.

This research investigates a quantitative framework for evaluating and ranking the S&P 500 universe using multiple technical and fundamental factors. The study outlines a systematic approach to standardizing diverse data points into a cohesive scoring model, which allows for direct comparison between stocks of different sectors. By employing a rolling window framework, the author demonstrates how factor-scoring can be used to generate actionable signals and manage volatility in large-cap portfolios. The study highlights the necessity of dynamic thresholding, where the criteria for a "high" or "low" score adjust based on broader market conditions to maintain the relevance of the rankings. This source is highly credible as it was published in the proceedings of a 2025 international finance conference. It is directly applicable to the development of the 13-metric engine, providing a mathematical basis for how to weigh and aggregate individual metrics into a final rank. The discussion on standardizing vectors ensures that the project's scoring logic remains statistically sound when comparing growth-heavy technology stocks against more stable industrial equities.

[7] Robert P. Schumaker and Hsinchun Chen. Textual analysis of stock market prediction using breaking financial news: The azfin text system. *ACM Transactions on Information Systems*, 27(2):12:1–12:19, 2009. Accessed: 2026-01-27.

Schumaker and Chen present a system for automating the analysis of financial news to predict stock prices. While my project will focus largely on numerical ratios, I also have to score qualitative metrics like Leadership and Culture. This paper is essential because it demonstrates how to treat unstructured text as data. The authors use a machine learning approach to tag and classify financial terms, which is the same logic I will need to implement if I want my web app to automatically score the News/Catalyst portion of my dashboard.

[8] Gautam Solaimalai. Serverless architecture for real-time stock market data analytics in cloud environments. *International Journal of Computer Applications*, 186(75):9–15, 2025.

This article presents a technical architecture for processing high-velocity stock market data using serverless cloud computing. The proposed system utilizes a microservices design to handle the extraction and transformation of stock signals, providing a scalable solution for real-time financial analytics. The author details the use of distributed event processing to manage high volumes of concurrent requests while maintaining low latency. The paper also evaluates the cost-effectiveness of serverless models compared to traditional infrastructures for data-heavy applications. Published in a peer-reviewed computer science journal in 2025, this source is a credible and highly relevant guide for the backend of the stock scoring engine. It provides the technical logic needed to fetch and process data for all 500 companies in the S&P 500 simultaneously. The serverless approach ensures that the application remains responsive even as more complex metrics are added to the 13-metric system. The focus on low-latency data pipelines is especially useful for the News/Catalyst and Price Action components of the project, where data speed is vital for accurate scoring.

[9] S&P Dow Jones Indices. S&p u.s. indices methodology. Technical report, McGraw Hill Financial, February 2016. Accessed: 2026-01-27.

This official methodology document from S&P Dow Jones Indices outlines the eligibility criteria and maintenance rules for the S&P 500 index. It defines the universe of stocks my application will rank. Understanding the divisor-based weighting and float-adjustment ensures that my app's data filtering logic remains consistent with industry standards.